

Non-Infectious Pathology in Asian Elephants

Comprehensive Study Notes with Slide Images

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Training Program: Essentials for Mortality Investigation of Asian Elephant Chhattisgarh Forest Department · 5–6 June 2026, Raigarh · Technical Support: WII Dehradun

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06-06-2026

NON INFECTIOUS PATHOLOGY IN ASIAN ELEPHANTS



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ELEPHANT POISONING IN INDIA

Poisoning is a severe threat to pachyderms –

In India and across Africa, these incidents primarily stem from **human-wildlife conflict** and **illegal poaching**.

A **pachyderm** is a large, thick-skinned mammal. The term is most commonly used for elephants, rhinoceroses, and hippopotamuses.

Poisoning intentionally via illegal poaching and human-elephant conflict retaliation, or accidentally. In India, unnatural causes like electrocution and poisoning claim hundreds of lives.

1. Introduction — Non-Infectious Pathology & Elephant Poisoning

1.1 Scope

Non-infectious pathology = disease/death NOT caused by transmissible pathogens. In Asian elephants these causes are overwhelmingly ANTHROPOGENIC and are the leading cause of unnatural elephant mortality in India: poisoning, electrocution, metabolic/reproductive disorders, and nutritional deficiency.

1.2 Elephant Poisoning in India

Poisoning is a severe threat to pachyderms. Incidents primarily stem from HUMAN-WILDLIFE CONFLICT and ILLEGAL POACHING — intentionally (poaching, retaliation) or accidentally (agrochemicals, contaminated crops/water).

- Elephants are BULK FORAGERS — 150–300 kg food + 100–200 L water/day → highly susceptible to environmental toxins
- Unnatural causes (electrocution, poisoning) claim hundreds of elephant lives yearly in India

KEY: Non-infectious elephant mortality is largely PREVENTABLE — it results from human activity. Correct cause-of-death identification is essential for legal action and prevention.

2. Poisoning Statistics & Categories of Toxicity

06-06-2026

ELEPHANT POISONING IN INDIA		
Elephant deaths caused by poisoning since 1997 in different states of India		
STATE	YEAR CASES	
Arunachal Pradesh	1997-2005	02
Assam	1997-2005	29
Jharkhand	2002-2003	02
Karnataka	1997-2005	02
Kerala	1997-2003	02
Meghalaya	1997-2005	01
Orissa	1997-2003	03
Tamil Nadu	1997-2005	01
Uttar Pradesh	1997-2004	02
Uttaranchal	2001-2004	02
West Bengal	1997-2004	03
TOTAL		49

Source: WTI elephant mortality database

Elephant poisoning is often due to anthropogenic (human-caused) events and natural toxicities. Elephant toxicity primarily falls into three categories:
I. Agricultural / plant mycotoxins.
II. Human-wildlife conflict chemicals.
III. Environmental toxins.
Elephants are bulk foragers, high quantity of daily food and water intake makes them highly susceptible.
1. Anthropogenic Poisons
Human-elephant conflict may result in to intentional or accidental poisonings, including:
i. Agrochemicals - Pesticides: Elephants raiding crops / plantations - consume dangerous levels of agrochemicals, mostly deadly neurotoxins i.e. organophosphates (like chlorpyrifos) and neonicotinoids (like imidacloprid).
ii. Cyanide: Used in illegal poaching and mining operations, it can cause rapid, fatal hypoxia. Poachers - Natural salt licks with cyanide near waterholes
iii. Fertilizers/Heavy Metals: Accidental urea fertilizer toxicity or heavy metal contamination (such as arsenic and lead) can destroy gastrointestinal and renal integrity.

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Slide 2 — State-wise poisoning deaths (1997–2005, total 49); three categories of toxicity

2.1 Elephant Deaths by Poisoning, State-Wise (WTI Database)		
State	Years	Cases
Assam	1997–2005	29
Orissa	1997–2003	03
West Bengal	1997–2004	03
Arunachal Pradesh	1997–2005	02
Jharkhand	2002–2003	02
Karnataka	1997–2005	02
Kerala	1997–2003	02
Uttar Pradesh	1997–2004	02
Uttaranchal	2001–2004	02
Meghalaya	1997–2005	01
Tamil Nadu	1997–2005	01
TOTAL	1997–2005	49

- Assam = 29/49 (59%) — intense human-elephant conflict in tea-estate/agricultural landscapes
- Documented cases only — actual figures considered substantially higher (under-reporting)

2.2 Three Categories of Elephant Toxicity		
Category	Description	Examples
I. Agricultural / Plant Mycotoxins	Fungal toxins on raided crops	Cyclopiazonic acid (CPA) on Kodo millet; aflatoxins
II. Human-Wildlife Conflict Chemicals	Deliberate/accidental chemicals	Pesticides (organophosphates), cyanide, urea

III. Environmental Toxins	Naturally occurring toxins	Cyanobacteria (blue-green algae); toxic plants; heavy metals
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SUSCEPTIBILITY: Bulk foraging (150–300 kg/day) means even low-concentration toxins reach lethal total doses. Hindgut fermentation also releases toxins (e.g. HCN) over an extended period.

3. Anthropogenic Poisons

3.1 Agrochemicals — Pesticides (Neurotoxins)

- Crop-raiding elephants consume organophosphates (chlorpyrifos, monocrotophos), carbamates, neonicotinoids (imidacloprid)
- OP/Carbamate mechanism: inhibit acetylcholinesterase → acetylcholine accumulates → cholinergic crisis
- SLUDGE signs: Salivation, Lacrimation, Urination, Defecation, GI distress, Emesis; miosis; muscle fasciculation
- Death from respiratory failure; ANTIDOTE: Atropine + Pralidoxime (2-PAM, for OP only)

3.2 Cyanide

- Used in illegal poaching/mining → rapid fatal hypoxia; poachers lace salt licks near waterholes
- Binds cytochrome c oxidase → blocks mitochondrial oxygen use (see Section 5 for detail)

3.3 Fertilizers / Heavy Metals

- UREA toxicity → ammonia → neurological signs, death
- LEAD: encephalopathy, GI colic, kidney damage (paint, batteries, mining waste)
- ARSENIC: haemorrhagic enteritis, liver/kidney damage (pesticides, industrial effluent)

SAMPLING: gut contents, liver, kidney, brain, blood (frozen). OP/carbamate → blood AChE. Heavy metals → liver/kidney for AAS. Always collect suspected bait. Send to IVRI/State FSL with suspected-diagnosis note.

4. Mycotoxins & Biological Toxins

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2. Mycotoxins Biological Toxins

- I. Crop-Borne Toxins:** Elephants raiding farm fields are highly susceptible to mycotoxins like *Cyclopiazonic Acid (CPA)*. This fungal toxin, which often infects crops like kodo millet, can cause acute vascular damage and liver/kidney necrosis.
The recent case seen in Madhya Pradesh - 13 elephants of a herd raided the Kodo millet crops near the Bandhavgarh Tiger Reserve.
Ten elephants died. Forensic postmortem and lab. investigations revealed that the deaths were caused by accidental toxicity from eating crops infected with fungal mycotoxins.
I have seen / treated two cases in KTR during 1997-98
- II. Cyanotoxins / Cyanobacteria:**
Toxic blue-green algae blooms in stagnant, seasonal water sources produce potent neurotoxins and hepatotoxins that have caused fatal die-offs. Mass deaths have occasionally been linked to toxic algal blooms (**blue-green algae / cyanobacteria**) in watering holes, particularly during hot, dry seasons.
- III. Toxic Plants:** Elephants are generally adept at avoiding toxic vegetation, ingestion of plants like **cleander** or **certain weeds** can result in lethal cardiac or neurological effects.
some local toxic plants (such as those containing calcium oxalate crystals) can cause localized severe irritation, swelling, and systemic distress if consumed in large quantities.

The recent case in Madhya Pradesh - 13 elephants of a herd raided the Kodo millet crops near the Bandhavgarh Tiger Reserve.
Ten elephants died. Forensic and autopsy investigations revealed that the deaths were caused by accidental toxicity from eating crops infected with poisonous fungal mycotoxins.
I have seen / treated two cases - in camp elephants of KTR during 1997-98

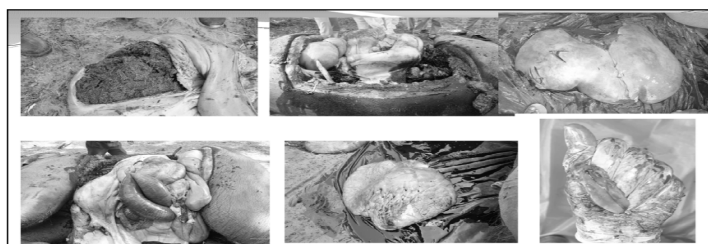
Kodo poisoning in elephants is caused by contaminated kodo millet crop with **mycotoxins**, produced by fungi primarily **cyclopiazonic acid (CPA)** (*Aspergillus* and *Penicillium* species)
This contamination typically occurs when heavy or unseasonal rainfall coincides with the grain's maturation or harvest, creating moist conditions favorable for fungal growth.

Kodo millet toxicity management- by combination of preventative measures in cultivation, storage, and processing. Biocontrol agents, such as non-toxigenic fungal strains, can help control fungal growth and mycotoxin production.
Good agricultural practices and proper post-harvest handling of kodo, proper drying and airtight storage.

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Slide 3 — Mycotoxins (CPA/Kodo); cyanotoxins; toxic plants; Bandhavgarh Kodo case

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Slide 4 — Field photographs: raided Kodo millet crop; necropsy organ findings

4.1 Crop-Borne Mycotoxins — Kodo Millet (Cyclopiazonic Acid)

Elephants raiding farm fields are highly susceptible to mycotoxins like Cyclopiazonic Acid (CPA) — a fungal toxin (*Aspergillus*, *Penicillium*) that infects Kodo millet (*Paspalum scrobiculatum*) and causes acute vascular damage and liver/kidney necrosis. Contamination occurs when heavy/unseasonal rainfall coincides with grain maturation.

CASE — Bandhavgarh Tiger Reserve (MP): A herd of 13 elephants raided Kodo millet crops; TEN ELEPHANTS DIED. Forensic PM + lab confirmed accidental mycotoxin (CPA) toxicity. Dr. Shrivastava: "I have seen/treated two cases in KTR during 1997–98."

- Management: preventive cultivation/storage/processing; biocontrol (non-toxigenic fungal strains); proper drying + airtight storage of Kodo

4.2 Cyanotoxins / Cyanobacteria (Blue-Green Algae)

- Toxic blue-green algae blooms in stagnant seasonal water → neurotoxins + hepatotoxins → fatal die-offs
- Common in hot, dry seasons when water sources shrink/concentrate
- Microcystins (hepatotoxins, Microcystis) → liver necrosis; Anatoxin-a/Saxitoxins (neurotoxins) → respiratory paralysis
- 2020 Botswana (Okavango) African elephant mass die-off attributed to cyanobacterial neurotoxins

4.3 Toxic Plants

- Elephants generally avoid toxic vegetation, but oleander (cardiac glycosides) → fatal arrhythmia
- Calcium oxalate-containing plants → severe oral/GI irritation, swelling, systemic distress

5. HCN (Cyanide) Poisoning from Sorghum

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HCN (Hydrocyanic acid or prussic acid) poisoning in elephants consuming sorghum. Occurs when the plant is stressed (drought, frost, or trampling). Ingestion leads to rapid cellular asphyxiation.

The Mechanism
Sorghum species (*Sorghum bicolor*) contain a cyanogenic glucoside called dhurrin. When the plant is chewed or its cells are ruptured, enzymes in the plant mix with dhurrin, releasing hydrogen cyanide. The toxin rapidly absorbed into bloodstream - blocks cellular oxygen use - death by asphyxiation.

Signs and Symptoms

- **Rapid onset:** Symptoms appear within minutes to a few hours of consumption.
- **Behavioural changes:** Laboured and rapid breathing, staggering gait, muscle tremors, convulsions, recumbency, and sudden death.
- **Physical signs:** A distinct bitter almond smell, excessive salivation, and **bright cherry-red blood** and mucous membranes (unlike nitrate poisoning, which causes brown blood).

Risk Factors for Elephants

- **Plant Height/Maturity:** Young, dark green sorghum regrowth (usually less than 50 cm tall) holds the highest concentration of toxins.
- **Environmental Stress:** Drought-stressed, wilted, or frost-damaged sorghum plants exhibit toxicity.
- **Digestive Process:** Elephants are hindgut fermenters, large quantities of fresh, young, or stressed sorghum fodder can still quickly overwhelm their metabolic detoxification capacities.

Management and First Aid

Remove the food source: Move the herd away from the sorghum field immediately.

Antidote Therapy: Administration of specific cyanide antidote kits. (**Sodium nitrite** and **sodium thiosulfate**). Sodium nitrite forces the conversion of hemoglobin to methemoglobin (which binds with cyanide), and sodium thiosulfate converts the cyanide into a harmless thiocyanate that can be excreted in urine.

Antidote & Treatment

- **Sodium Nitrite:** Administered IV (e.g., 1gm to 3gm in 50ml distilled water) to convert hemoglobin into methemoglobin. This binds with the free cyanide to form cyanmethemoglobin.
- **Sodium Thiosulphate:** Followed immediately by an IV injection of sodium thiosulphate. This converts the cyanide into non-toxic thiocyanate, which is then safely excreted in urine.

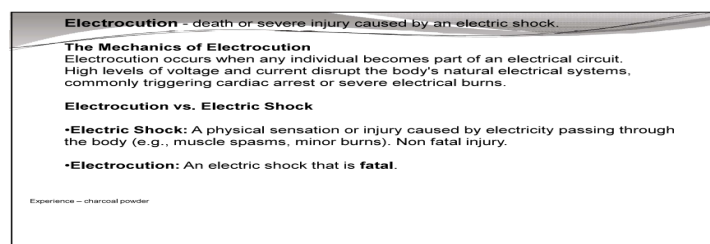
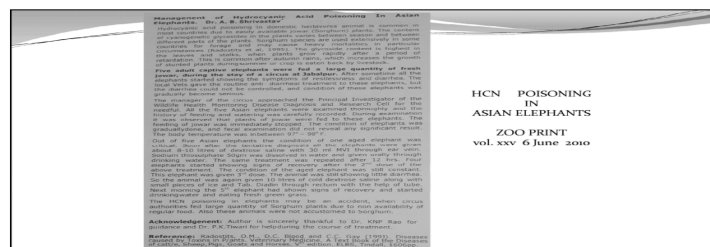
Prevention

- **Avoid grazing/feeding young plants:** Do not graze livestock on sorghum that is less than 18 to 24 inches tall (under 7 weeks of age), as this is when concentrations are highest.
- **Wait after weather stress:** Avoid grazing immediately after a drought ends or a frost occurs, as stress spikes the *dhurrin* content.
- **Hay vs. Pasture:** Sorghum hay can be safer than fresh pasture as the curing process reduces HCN levels by roughly half, but testing forage before feeding is highly advised

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Slide 5 — HCN poisoning: mechanism, signs, risk factors, management & antidote

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Slide 6 — HCN case report (Zoo Print 2010); Electrocution introduction

5.1 Mechanism

HCN (prussic acid) poisoning occurs when elephants consume STRESSED sorghum (drought, frost, trampling) → rapid cellular asphyxiation. Sorghum (*Sorghum bicolor*) contains the cyanogenic glucoside DHURRIN. When chewed, plant enzymes mix with dhurrin → release hydrogen cyanide → binds cytochrome c oxidase → cells cannot use oxygen → histotoxic hypoxia → death.

5.2 Signs & Symptoms

- **RAPID ONSET:** minutes to a few hours after consumption
- Laboured/rapid breathing, staggering gait, muscle tremors, convulsions, recumbency, sudden death
- **BITTER ALMOND** smell; excessive salivation; **BRIGHT CHERRY-RED** blood and mucous membranes
- **DISTINCTION:** cyanide → cherry-red blood; nitrate poisoning → brown (chocolate) blood

5.3 Risk Factors

Risk Factor	Detail
Plant height/maturity	Young dark-green regrowth (<50 cm tall) = highest toxin concentration
Environmental stress	Drought-stressed, wilted, or frost-damaged plants are most toxic
Digestive process	Hindgut fermenters; large quantities overwhelm detoxification capacity

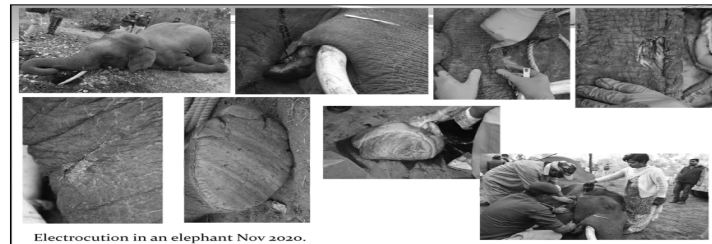
5.4 Antidote & Prevention

- Remove food source immediately; move herd away from sorghum field
- **SODIUM NITRITE IV** (1–3 gm in 50 ml distilled water) → forms methaemoglobin → binds cyanide as cyanmethaemoglobin
- **SODIUM THIOSULPHATE IV** → converts cyanide to non-toxic thiocyanate → excreted in urine
- Prevention: avoid young plants (<45–60 cm); wait after drought/frost; hay safer than fresh pasture (test forage)

CASE — Circus elephants, Jabalpur (Zoo Print 2010, Dr. Shrivastava): 5 adults fed fresh Jowar (unaccustomed) → restlessness, diarrhoea, subnormal temp 97–98°F. Treated: dextrose saline + MVI; sodium thiosulphate 50 gm oral, repeated 12 hrs; cold saline + ice + Diadin per rectum. Aged elephant recovered next morning.

6. Electrocutation in Elephants

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Pyometra

It is a severe, life-threatening uterine infection characterized by the accumulation of pus and severe inflammation in the uterus. It is most common in aging, nulliparous (having never given birth) female elephants, often linked to prolonged hormonal fluctuations and underlying reproductive issues.

Causes & Pathophysiology

- **Hormonal Changes:** Fluctuating progesterone levels and an abnormally thickened uterine lining create an environment favourable for bacterial growth
- **Bacterial Infection:** Bacteria like *E. coli* or *Enterococcus faecium*. Suppurative inflammation
- **Pre-existing Conditions:** Cystic endometrial hyperplasia or benign tumors (leiomyomas) frequently complicate the reproductive tract of older captive elephants and trigger pyometra

Treatment :

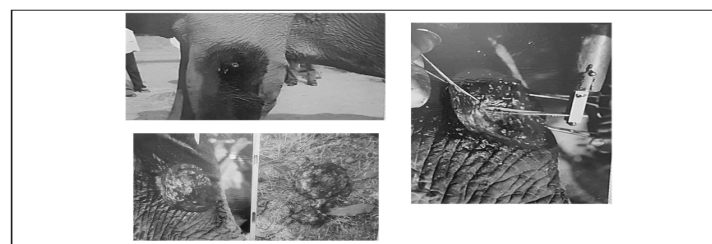
- Hormonal Downregulation:** The use of GnRH vaccines (like Improvac) or Deslorelin implants to suppress reproductive cycles and shrink the affected uterine tissue.
- Aggressive Antibiotic Therapy:** Broad-spectrum systemic antibiotics targeting specific pathogens.
- Uterine Lavage:** Flushing the uterus.
- Practiced - Pyrogenium 1000 + Heper sulf 1000 + Secele 1000**

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Slide 7 — Electrocution in an elephant (Nov 2020): field photographs

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Slide 9 — Electrocution case: carcass handling and postmortem photographs

6.1 Definition & Mechanics

Electrocution = death/severe injury from electric shock; occurs when the animal becomes part of an electrical circuit. High voltage/current disrupts the body's electrical systems → cardiac arrest or severe electrical burns.

Term	Definition
Electric Shock	Physical sensation/injury from electricity (muscle spasm, minor burns) — NON-FATAL
Electrocution	An electric shock that is FATAL

6.2 Context in India

- LEADING cause of unnatural elephant death in India (~70–75% of unnatural deaths)
- Sources: illegal low live wires around crops; sagging 11 kV power lines; solar-fence tampering
- Both intentional (poaching/retaliation) and accidental (sagging legal lines)

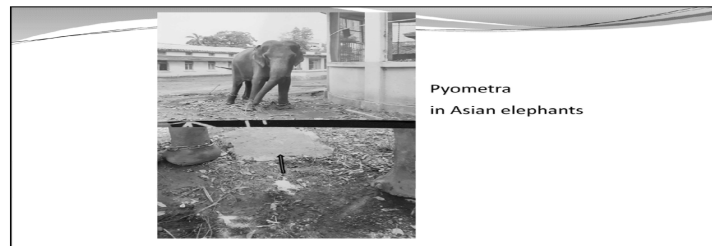
6.3 Postmortem Findings

- Electrical BURN MARKS (Joule burns) at contact point — trunk tip, ear, shoulder, feet; charring, crater lesions
- Subcutaneous/intramuscular haemorrhage along current path; entry & exit current marks
- Internal: pulmonary congestion/oedema; cardiac petechiae; generalised venous congestion
- Carcass near electrical source; Dr. Shrivastava note: "charcoal powder" appearance of charred contact tissue

INVESTIGATION: photograph scene (wires, poles, sag height, distance); burn marks with scale bar; skin histology (coagulative necrosis, epidermal nuclear streaming); coordinate with electricity board/police (punishable under WPA 1972 & Electricity Act); distinguish lightning (fern-pattern singeing, multiple animals).

7. Pyometra

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Hyperthermia Life-threatening – caused by high environmental temp, b a lack of access to water or shade, massive body size and lack of sweat glands.

Why Elephants Are Prone to Overheating

- **The Square-Cube Law:** As body volume increases, heat production more on skin surface. Massive size makes it good to retain heat, but poor at losing it.
- **Lack of Sweat Glands:** Elephants do not have traditional sweat glands, distributed across their bodies. They only have pores located between their toes, relying on "transepidermal water loss" to sweat through their skin.
- **Metabolic Heat:** More Physical activity, (continuous walking hot sunlight without rest and water) generates high metabolic heat that can easily trigger hyperthermia if the animal cannot cool down.

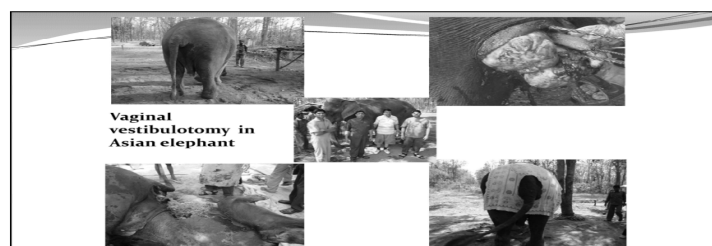
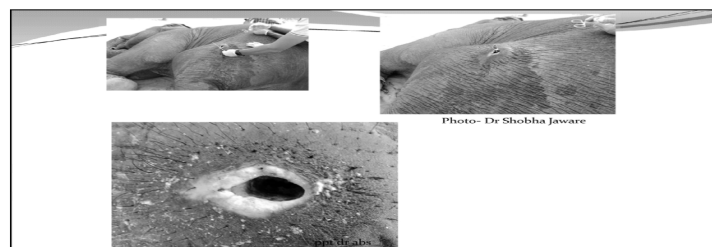
Natural Cooling Mechanisms to prevent heat stress

- **The Ears:** Elephant ears act as giant thermal windows. By flapping their ears, they cool the blood circulating through them, reducing their overall body temperature significantly.
- **Wallowing & Spraying:** Bathing in mud and water is vital. Spraying own saliva. Water evaporates provides an intensive cooling effect, and the residual mud protects their skin from solar radiation.
- **Skin Permeability:** Elephant's hide becomes more permeable in hot weather, allowing them to lose moisture and cool down faster through evaporation, provided they have access to drinking water.

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Slide 8 — Pyometra in Asian elephants (clinical photograph; purulent discharge)

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Slide 10 — Vaginal vestibulotomy in Asian elephant: surgical drainage

7.1 Definition

Pyometra = severe, life-threatening uterine infection with accumulation of PUS and severe inflammation. Most common in AGING, NULLIPAROUS (never-bred) females, linked to prolonged hormonal fluctuation and reproductive issues.

7.2 Causes & Pathophysiology

- Hormonal: fluctuating progesterone + thickened endometrium favour bacterial growth

- Bacterial: E. coli or Enterococcus faecium → suppurative inflammation
- Pre-existing: Cystic Endometrial Hyperplasia (CEH) or leiomyomas in older captive elephants trigger pyometra
- Nulliparous state → repeated uninterrupted cycles → continuous progesterone → CEH-pyometra complex

7.3 Treatment

Approach	Detail
Hormonal downregulation	GnRH vaccines (Improvac) or Deslorelin implants to suppress cycles, shrink tissue
Antibiotic therapy	Broad-spectrum systemic antibiotics targeting cultured pathogens
Uterine lavage	Flushing the uterus to evacuate pus
Surgical drainage	Vaginal vestibulotomy (see slide 10)
Homeopathic (practised)	Pyrogenium 1000 + Hepar sulph 1000 + Secale 1000 (Dr. Shrivastava)

Pyometra is a reproductive EMERGENCY — uterine endotoxaemia can cause septic shock and death. Prevention in captive females: managed breeding or hormonal cycle suppression to break the CEH→pyometra cascade.

8. Hyperthermia & Heat Stress

8.1 Definition

Hyperthermia is a life-threatening condition from high environmental temperature, lack of access to water/shade, massive body size, and lack of sweat glands — a serious mortality concern during Indian summers.

8.2 Why Elephants Overheat

- **SQUARE-CUBE LAW:** heat production rises faster than skin surface area; massive size retains heat but loses it poorly
- **NO SWEAT GLANDS:** pores only between toes; rely on transepidermal water loss
- **METABOLIC HEAT:** continuous walking in hot sun without rest/water triggers hyperthermia

8.3 Natural Cooling Mechanisms

Mechanism	How It Works
The Ears	Giant thermal windows; flapping cools blood in dense ear vasculature
Wallowing & Spraying	Mud/water evaporation cools; residual mud blocks solar radiation; saliva spraying
Skin Permeability	Hide becomes more permeable in heat → faster evaporative cooling (needs drinking water)
Skin Wrinkles	Increase surface area; retain water/mud → prolong evaporative cooling

8.4 Prevention & Management

- Constant access to **SHADE** + clean **DRINKING WATER**, especially 11 AM–4 PM
- Provide wallowing pools/mud baths; avoid working elephants during peak heat
- Recognise signs: rapid breathing, intensified ear-flapping, lethargy, collapse
- Emergency cooling: continuous water spray (ears/head), shade, IV fluids

9. Iron Deficiency Anaemia

06-06-2026

Iron Deficiency Anaemia

Anaemia – qualitative and quantitative reduction quality of the blood.

Symptoms observed – pale m m, extremely weakness, unable to walk, very slow walk and other activity.

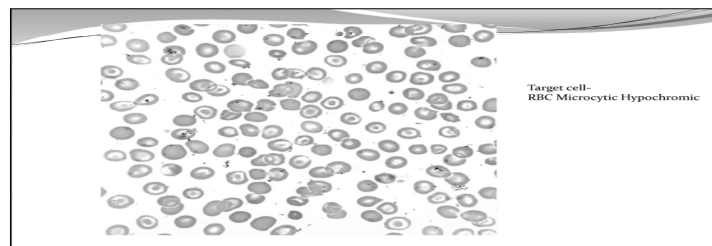
Diagnose- Blood examination

Low Hb

Blood smear – typical target cell (RBC)

Treatment:

1. Diet change –to full normal elephant diet soyabeans, horse gram, napier grass, bamboo, leaves, grains, fruits
2.
 - i. Iron supplements – Jaggery, honey
 - ii. B12 injections
 - iii. Multivitamins, M mixture



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Slide 11 — Blood smear: microcytic hypochromic target cells (codocytes)

9.1 Definition & Signs

Anaemia = qualitative and quantitative reduction in blood quality / oxygen-carrying capacity. Iron deficiency → microcytic, hypochromic RBCs. Signs: pale mucous membranes, extreme weakness, unable to walk, very slow walk and reduced activity.

9.2 Diagnosis

- Blood examination: LOW Haemoglobin (Hb)
- Blood smear: typical TARGET CELLS (codocytes) — MICROCYTIC, HYPOCHROMIC RBCs
- Reference Hb (Indian captive elephants): ~8.6–16.8 g/dL; below range confirms anaemia

9.3 Treatment

- Diet change to full normal elephant diet: soybeans, horse gram, Napier grass, bamboo, leaves, grains, fruits
- Iron supplements: Jaggery (gur), honey
- Vitamin B12 injections; Multivitamins + mineral mixture

INVESTIGATION: Iron deficiency anaemia is often **SECONDARY** — investigate chronic blood loss (ticks, lice, GI helminths), nutritional inadequacy, or chronic disease. Run a parasitological exam alongside iron supplementation to prevent relapse.

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Prof. Dr. A B Shrivastava, NDVSU Jabalpur | Presentation: 06-06-2026 | Training Program: Mortality Investigation of Asian Elephant, Raigarh, Chhattisgarh | Notes elaborated with peer-reviewed literature